Errata

Statistics for Chemical Engineers: From Data to Models to Decisions

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This list outlines errata in the 1st edition of the book as printed by Cambridge University Press (2025); section, example, and equation numbers refer to the printed book. All corrections listed here are incorporated in the 2nd edition. To report an error, please use the form on the book website.

Book

Location	**Error**	**Correction**
Eq. (1.5)	$\frac{\partial \log K_{eq}}{\partial T} = -\frac{\Delta H_0}{R \cdot T}$	$\frac{\partial \log K_{eq}}{\partial T} = \frac{\Delta H_0}{R \cdot T^2}$ (van 't Hoff equation)
Sections 1.2, 1.3, 2.1	Poisson domain $\{0, 1, 2, \ldots, \infty\}$	$\{0, 1, 2, \ldots\}$
Example 2.2	"…the machine has not failed in 79 out of 156 weeks."	"…the machine has failed in 79 out of 156 weeks." (consistent with $\hat{\mathbb{P}}(X = 0) = 79/156$, where $X = 0$ denotes failure)
Example 2.4	"…for a Bernoulli model, we must have that $\mathbb{P}(X = 0) = p$ and $\mathbb{P}(X = 1) = 1 - p$."	"…we must have that $\mathbb{P}(X = 0) = 1 - p$ and $\mathbb{P}(X = 1) = p$."
Eq. (2.14)	$\mathbb{V}_X = n \cdot p \cdot (1 - p) \cdot \frac{N_t - n}{n - 1}$	$\mathbb{V}_X = n \cdot p \cdot (1 - p) \cdot \frac{N_t - n}{N_t - 1}$
Example 2.6	$\mathbb{P}(X = 5) = f(5) = 0.1312$	$\mathbb{P}(X = 5) = f(5) = 0.0312$
Example 2.6	$X \sim \mathcal{H}(N_y, N_t, p, n)$	$X \sim \mathcal{H}(N_y, N_t, n, p)$ (same parameter order as in the definition of the hypergeometric RV)
Discussion of Figure 2.5	"the pdf of the RV $\mathcal{H}(\mathbf{y}, N, n, p)$ for $N = 1000$, …"	"the pdf of the RV $\mathcal{H}(N_y, N_t, n, p)$ for $N_t = 1000$, …"

Location	Error	Correction
Section 2.2	"…will shrink/damp uncertainty if $b < 1$ and will magnify uncertainty if $b > 1$."	"…if $\|b\| < 1$ …if $\|b\| > 1$."
Exercise 2.6	"…estimated to be 15 ± 1 MWh. The daily energy demand is assumed to be Gaussian and the confidence intervals correspond to a probability level of 95%." The meaning of ± 1 is ambiguous (it can be read as a standard deviation of 1 MWh).	"…estimated to be 15 ± 1 MWh, where ± 1 MWh is the half-width of the 95% confidence interval (i.e., the 95% confidence interval of the daily energy demand is $[14, 16]$ MWh). The daily energy demand is assumed to be Gaussian."
Example 3.2	$(1 - 0.36) \cdot (1 - 0.90) = 0.064$	$(1 - 0.37) \cdot (1 - 0.90) = 0.060$
Eq. (3.29)	Last term: $+\mathbb{P}(X_1 \leq x_1 + \Delta x_1 \ \& \ X_2 \leq x_2)$	$-\mathbb{P}(X_1 \leq x_1 + \Delta x_1 \ \& \ X_2 \leq x_2)$
Example 3.16	$\Sigma_{2\|1} = 0.015$; "relative reduction of about 25%" (and of 21% for $\Sigma_{1\|2}$)	$\Sigma_{2\|1} = 0.016$; relative reduction of about 22% in both cases
Eq. (3.114)	$f(x_1, x_2) = \dfrac{1}{2\pi\sqrt{\Sigma_{11}\Sigma_{22}}} \exp(-E(x_1, x_2))$	$f(x_1, x_2) = \dfrac{1}{2\pi\sqrt{(1-\rho^2)\Sigma_{11}\Sigma_{22}}} \exp(-E(x_1, x_2))$
Example 4.1	"From the definition of the first central moment, we have $\mathbb{CMO}_1(\theta) = \mathbb{E}[(X - \mathbb{E}_X)] = \mathbb{E}_X - \mu = 0$."	"From the definition of the first raw moment, we have $\mathbb{RMO}_1(\theta) = \mathbb{E}_X = \mu$."
Example 4.2	$\hat{\mathbb{V}}_X = \beta^2 \left(\Gamma(1 + 2/\xi) + \Gamma(1 + 1/\xi)^2\right)$	$\hat{\mathbb{V}}_X = \beta^2 \left(\Gamma(1 + 2/\xi) - \Gamma(1 + 1/\xi)^2\right)$
Examples 4.6 and 4.7	$X \sim$ Weibull$(2, 1)$	$X \sim$ Weibull$(1, 2)$, that is, shape $\xi = 1$ and scale $\beta = 2$ in the notation Weibull(ξ, β) (this is consistent with $\mathbb{E}[X] = 2$ and $\mathbb{V}[X] = 4$ reported in Example 4.7)
Section 4.3	"…can achieved by using a powerful result…"	"…can be achieved by using a powerful result…"
Section 5.1	"…they will sit at a shaper minimum…"	"…they will sit at a sharper minimum…"
Section 5.3	"…θ are the kinetic constraints."	"…θ are the kinetic constants."

Location	Error	Correction
Example 5.10	"…lands at a minimum point that has nonnegative physical values."	"…lands at a minimum point that has negative (nonphysical) values."
Below Eq. (5.115)	"This penalizes squared deviations from a reference value $\hat{\theta}$."	"This penalizes squared deviations from a reference value $\bar{\theta}$."
Section 5.4	"(we will see this is Section 5.5)"	"(we will see this in Section 5.5)"
Example 5.12	$SSE(\theta) + \frac{1}{2}\|\|\theta - \bar{\theta}\|\|_2^2$	$SSE(\theta) + \kappa \cdot \frac{1}{2}\|\|\theta - \bar{\theta}\|\|_2^2$ (with $\kappa = 1$, as stated in the example)
Example 5.13	$\hat{\theta} = (0.25, 0.06, 0.04, 0.11, 1.19)$	$\hat{\theta} = (1.25, 0.06, 0.04, 0.11, 1.19)$ (this coincides with the estimates obtained in Example 5.10)
Eq. (6.4)	$\hat{\text{Cov}}(X) = \mathbf{X}^T\mathbf{X}$	$\hat{\text{Cov}}(X) = \frac{1}{S}\mathbf{X}^T\mathbf{X}$
Eq. (6.42)	$g_j = e^{-j^2/\sigma^2}$	$g_j = e^{-j^2/(2\sigma^2)}$ (the weights reported in Examples 6.9 and 6.10 use this form)
Example 6.10	$g = (0.0003, 0.41, 0.41, 0.003)$	$g = (0.0003, 0.41, 0.41, 0.0003)$
Section 6.3, inverse FT	$f(t) = \int_{-\infty}^{\infty} \hat{f}(\omega) e^{i(2\pi\omega\tau)} d\omega$	$f(t) = \int_{-\infty}^{\infty} \hat{f}(\omega) e^{i(2\pi\omega t)} d\omega$
Eq. (6.76)	$g(x,y) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x^2+y^2}{2\sigma^2}}$	$g(x,y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$
Section 6.5	"…a Gaussian prior $\theta \sim \mathcal{N}(0, \lambda\mathbf{I})$"	"…a Gaussian prior $\theta \sim \mathcal{N}(0, \lambda^{-1}\mathbf{I})$"
Eq. (6.123)	$\ldots - \frac{1}{2}\mathbf{y}^T\mathbf{C}(\gamma)\mathbf{y}\ldots$	$\ldots - \frac{1}{2}\mathbf{y}^T\mathbf{C}(\gamma)^{-1}\mathbf{y}\ldots$
Eq. (6.124)	$\sigma^2(x) = c(x,\hat{\gamma}) - \mathbf{k}(x,\hat{\gamma})^T\mathbf{C}(\hat{\gamma})\mathbf{k}(x,\hat{\gamma})$	$\sigma^2(x) = c(x,\hat{\gamma}) - \mathbf{k}(x,\hat{\gamma})^T\mathbf{C}(\hat{\gamma})^{-1}\mathbf{k}(x,\hat{\gamma})$
Example 7.5	"Standard deviation: $\mathbb{SD}[Y(u)] = 19{,}284$ USD, $\mathbb{V}[Y(u')] = 79{,}288$ USD."	"Standard deviation: $\mathbb{SD}[Y(u)] = 19{,}284$ USD, $\mathbb{SD}[Y(u')] = 79{,}288$ USD."
Example 7.5	"…a risk-conscious DM would prefer u because it has a lower worst-case value…"	"…a risk-averse DM would prefer u because it has a lower worst-case value…"
Eq. (7.33)	$\min_{u\in\mathcal{U}} \mathbb{P}(\varphi(X,u) > y)$	$\min_{u\in\mathcal{U}} \mathbb{P}(\varphi(X,u) > \bar{y})$

Location	Error	Correction
Section 7.7	"…we augment our dataset $\mathcal{D}_{\ell+1} \leftarrow \mathcal{D}_\ell \cup (u_\omega, y_\omega)$."	"…we augment our dataset $\mathcal{D}_{\ell+1} \leftarrow \mathcal{D}_\ell \cup \{(u'_\omega, y'_\omega)\}$."

Code, data, and slides

Location	Error	Correction
Exercise 1.7, `clinical_largegroup.dat`	The data file for the large groups (100 people) is identical to `clinical_smallgroup.dat`; the script `ch1_clinical_trials.m` saved the small-group data to both files.	Corrected: the script now saves the large-group data to `clinical_largegroup.dat`, and the data file has been regenerated (100 people, five trials).
Example 3.20 (e-book link)	The link to `ch3_gaussian_samples_ellipse.m` points to a misspelled file name (`ch3_gaussian_samples_eliipse.m`).	Link corrected.
Example 6.3, `ch6_gibbs_pca_estimation_sparse.ipynb`	The Python notebook does not reproduce the eigenvector matrices W reported in the example.	The notebook now uses the same sparse PCA method as the book and reproduces the informative columns of W and $\tilde{W}$. The input matrix has rank 3, so the last three eigenvalues are zero and the corresponding columns of W are not unique (they can differ across numerical libraries).
Example 6.26, `ch6_gibbs_neuralnet.ipynb`	The Python notebook produces results that differ substantially from those reported in the example.	Fixed in the October 2026 version of the notebook: the NN now fits the data as well as the GP model (mean squared error of about 9×10^{-4} for both), as reported in the example.
Chapter 3 slides	Discrete domain $\mathcal{D} = \{(l_1, l_2), (l_1, u_2), (u_1, l_2), (u_2, l_2)\}$.	$\mathcal{D} = \{(l_1, l_2), (l_1, u_2), (u_1, l_2), (u_1, u_2)\}$ (this is correct in the printed book).

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